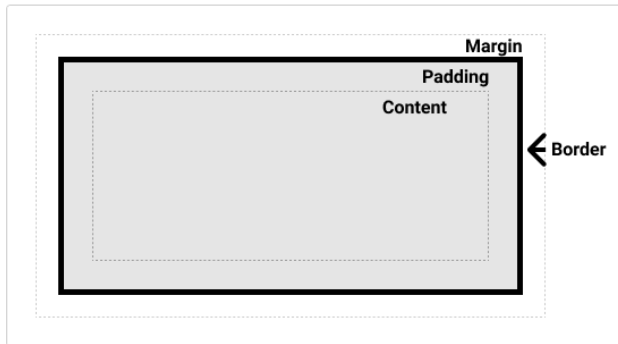


Preparation before next session

Read this **great article** about Box Model

[Article about Box Model](#)



Read this **great article** about CSS Units

[Article about CSS UNITS](#)

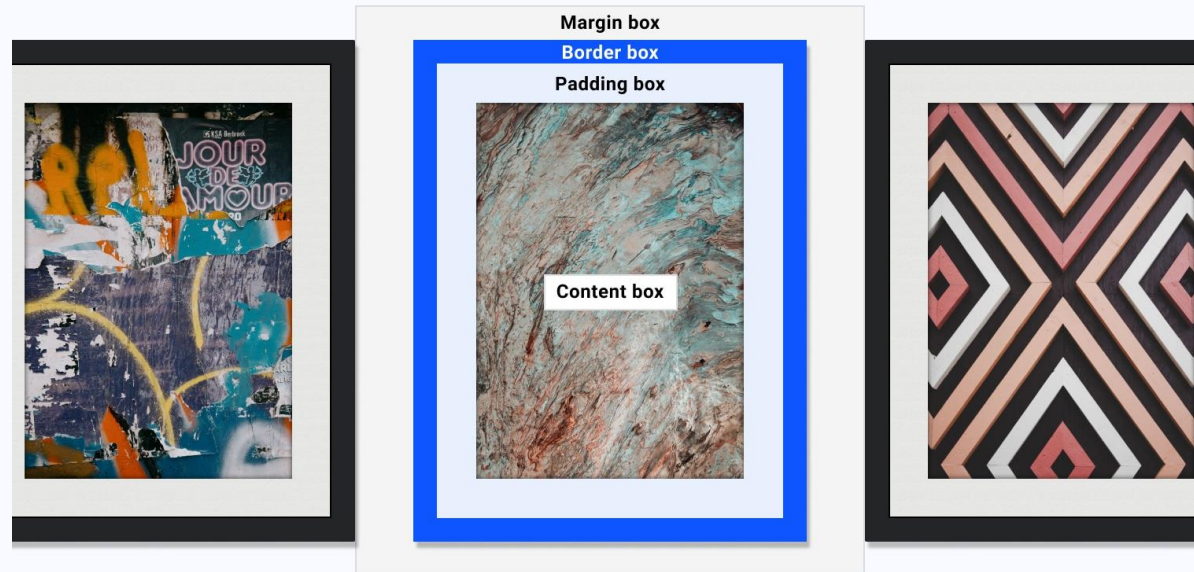
Watch this **great video** about Box Model

[video about Box Model](#)



WEB DESIGN

W3-S1 – Box Model & Units





Course Objectives



BOX MODEL

- ✓ Make the difference between **block** and **inline** elements
- ✓ Understand and manipulate the **Box Model** (Margin, Padding Border)
- ✓ Understand the **Box Sizing** modes to compute object dimensions

UNITS

- ✓ Understand the **main types of units** in web design

BLOCK VS INLINE

This is a block.

This is a block.

inline

inline

inline

This is also an inline.

Block VS Inline Elements

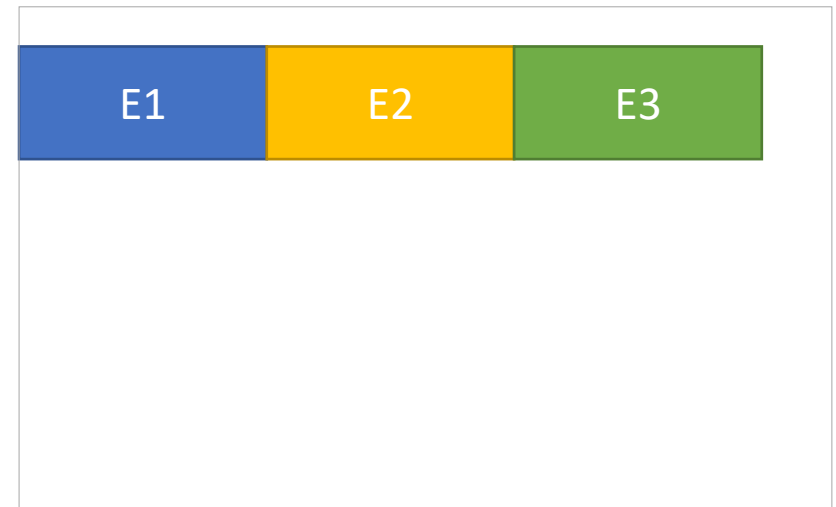
Block Elements

- ✓ Always starts on a **new line**
- ✓ Always takes up the **full width**



Inline Elements

- ✓ Stay on the **same line**
- ✓ Only takes up as much width as **necessary**



Block VS Inline Elements

Here are some block and inline elements

Block Elements

`<p>`
`<div>`

`<main>`
`<section>`
`<table>`
`<nav>`
`<ul>`
`<ol>`
`<li>`

Inline Elements

`<span>`
`<a>`

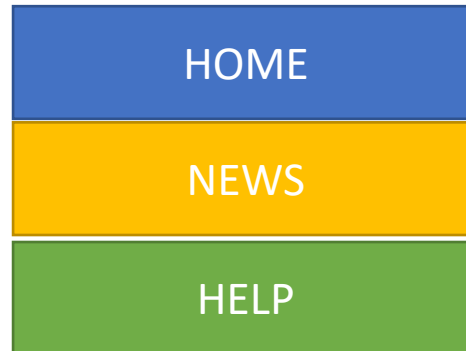
`<i>`
`<img>`
`<input>`
`<q>`

Change the Display property

You can change an element's display type using the display attribute

```
<ul>  
  <li>HOME  
  <li>NEWS  
  <li>HELP  
</ul>
```

```
li {  
  display: block  
}
```



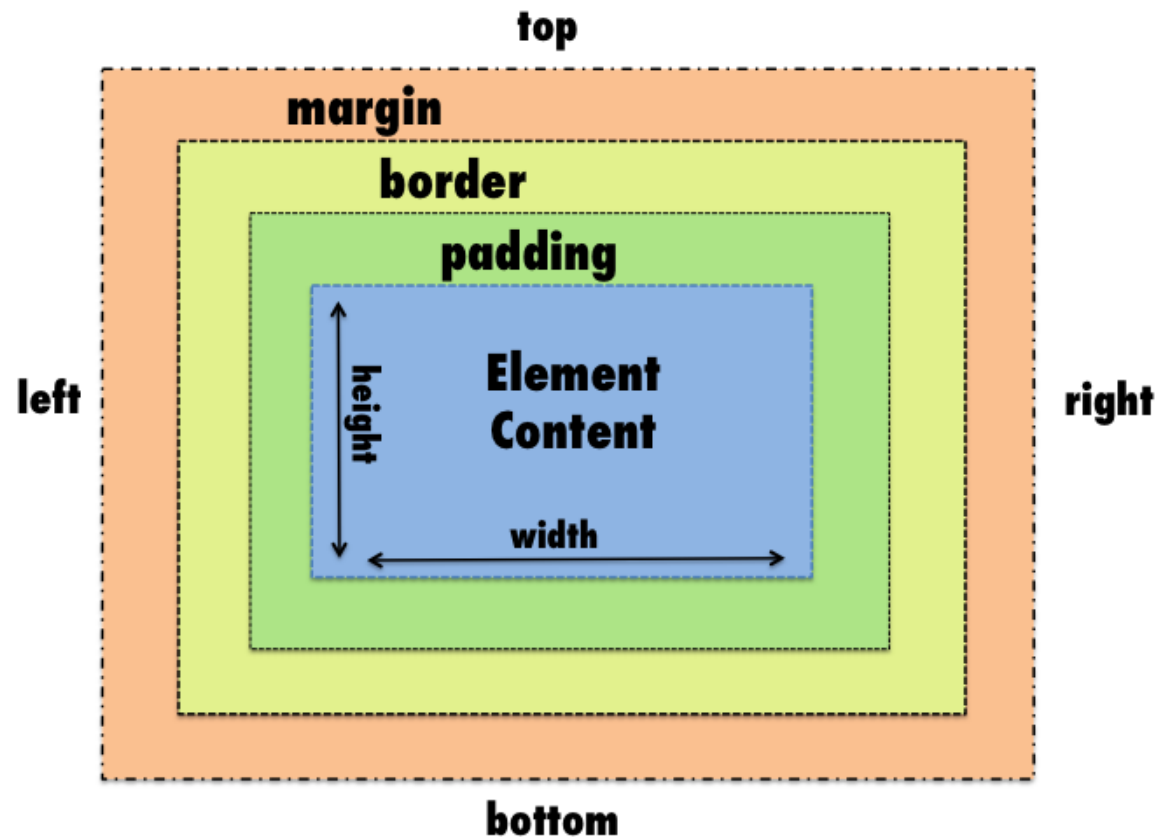
```
li {  
  display: inline  
}
```



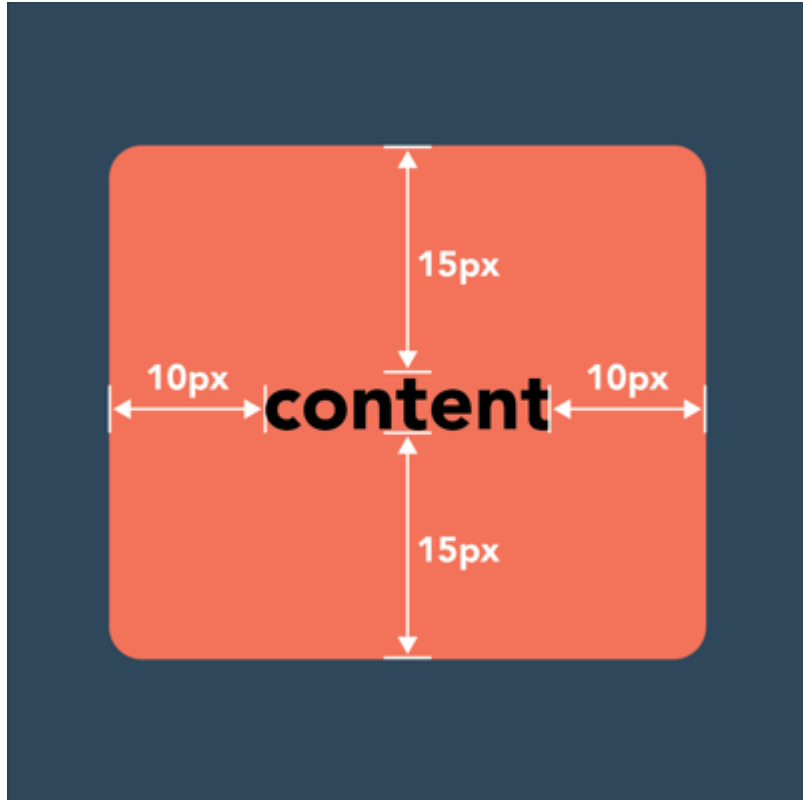
Create a horizontal menu by setting LI elements with an inline display

The box model

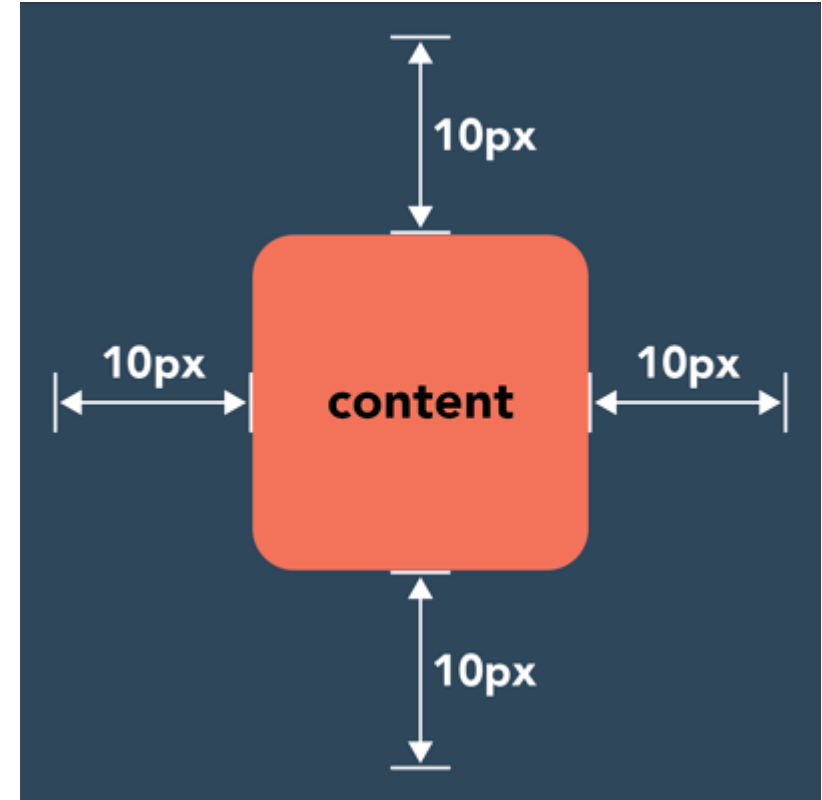
The CSS box model is a box that wraps around every HTML **element**. It consists of: **margins**, **borders**, **padding**, and the actual content



Which diagram refers to **padding**?

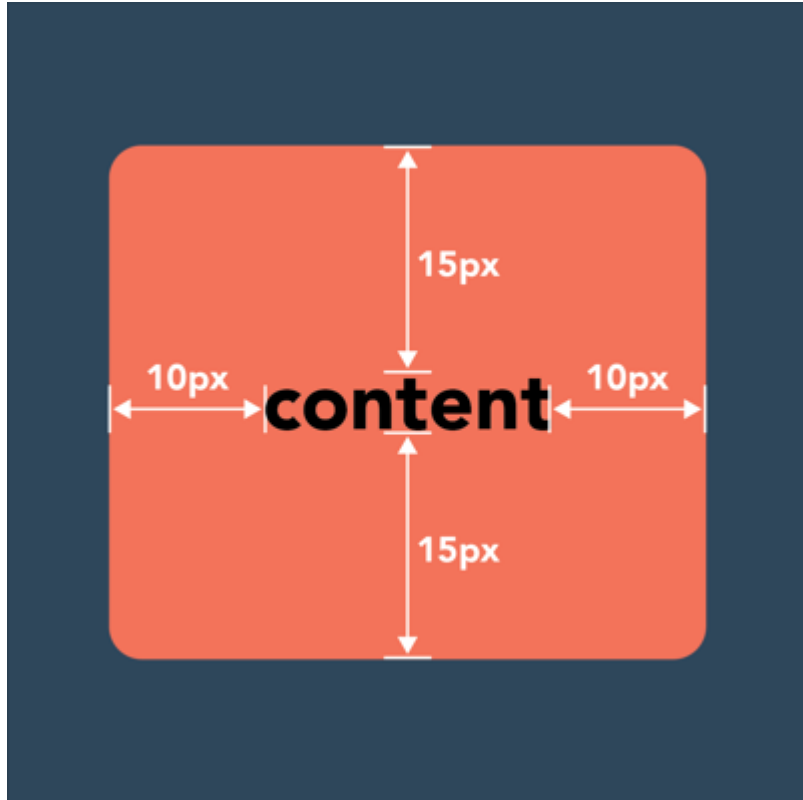


1

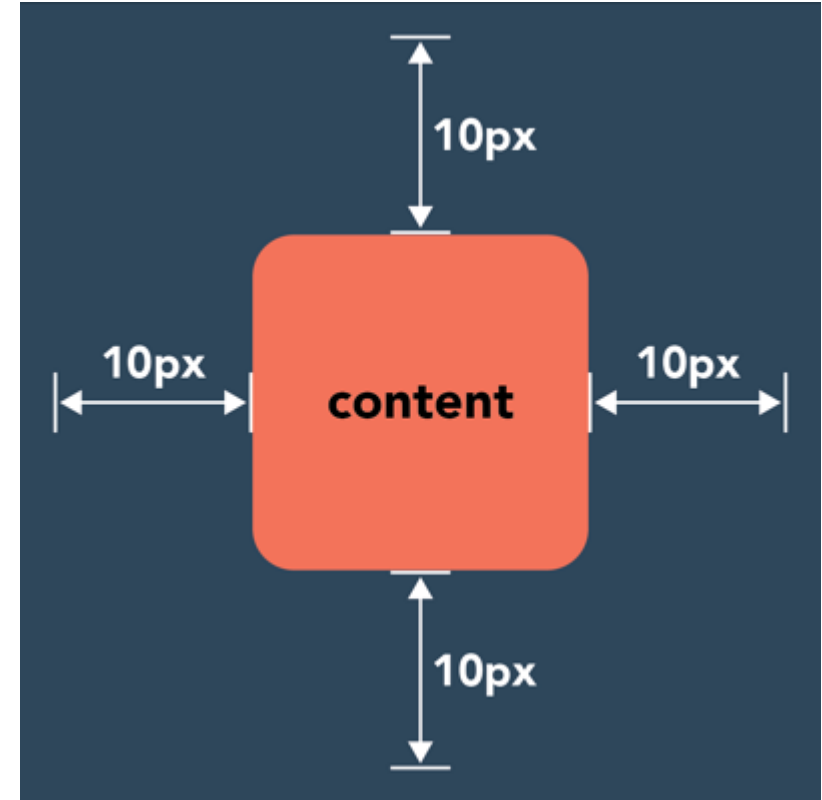


2

Which diagram refers to **padding**?



1



2



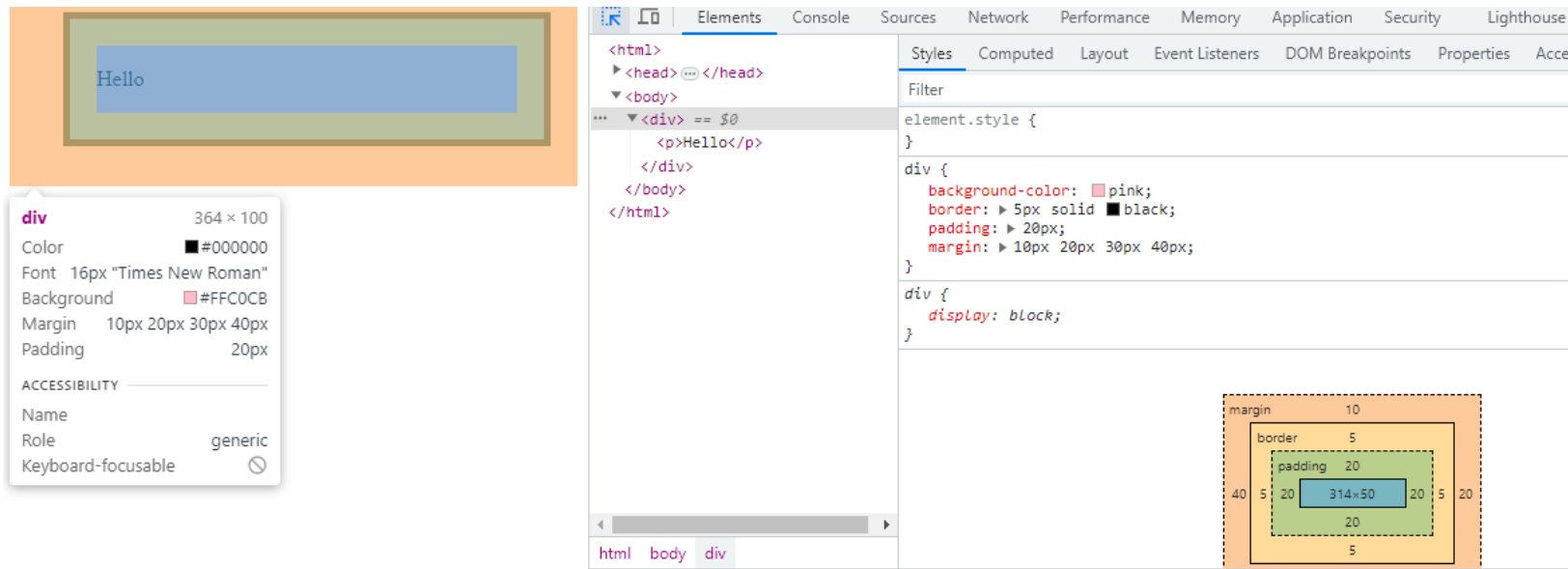
10 MIN



Activity 1

Look at the provided code and evaluate the margins, paddings and border of elements

- ✓ Open Dev Tools on Chrome
- ✓ Select an element
- ✓ Show the box model debugger



The box model debugger

The **box model** attributes

Margin

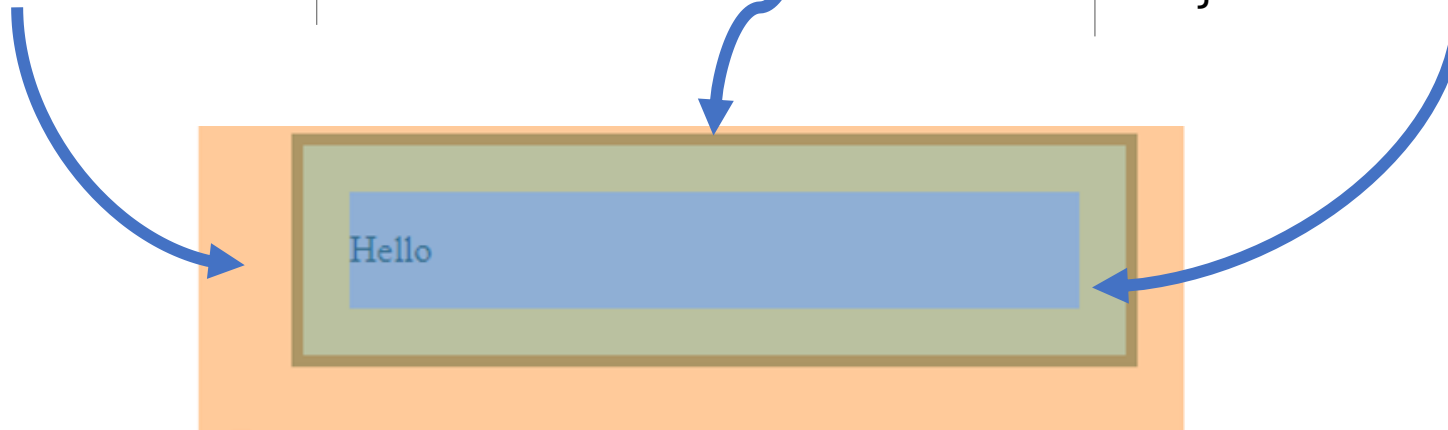
```
p {  
  margin-top: 100px;  
  margin-bottom: 100px;  
  margin-right: 150px;  
  margin-left: 80px;  
}
```

Border

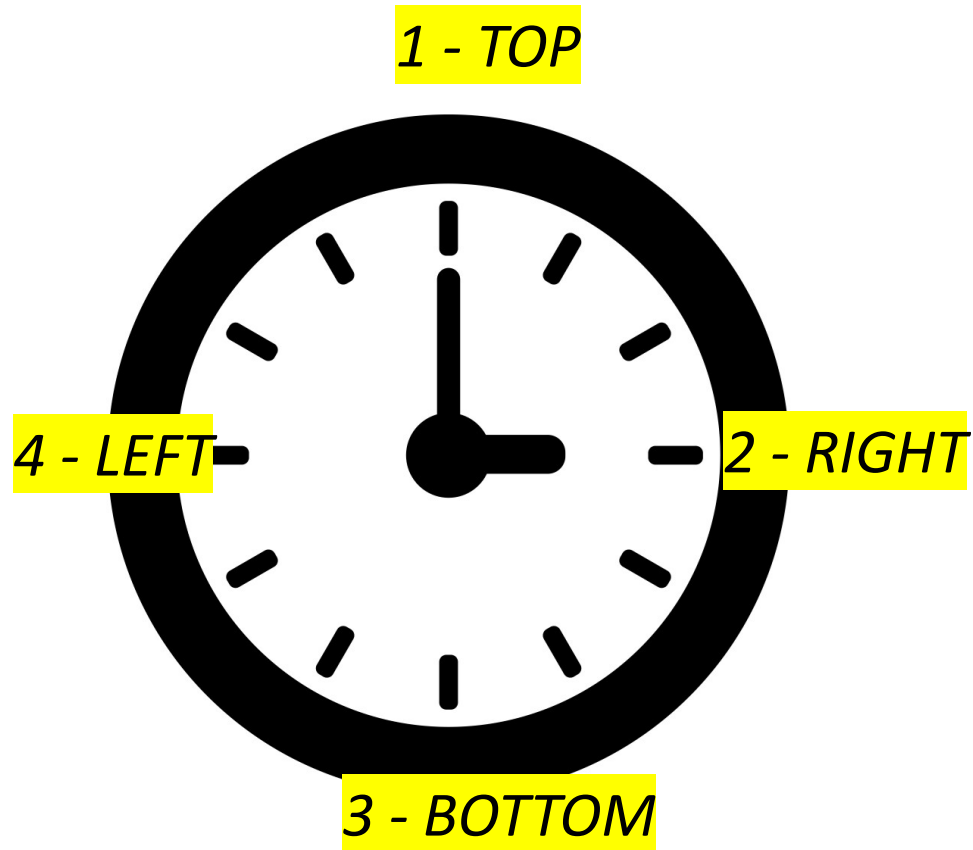
```
div {  
  border-style: solid;  
  border-width: 5px;  
}
```

Padding

```
div {  
  padding-top: 50px;  
  padding-right: 30px;  
  padding-bottom: 50px;  
  padding-left: 80px;  
}
```



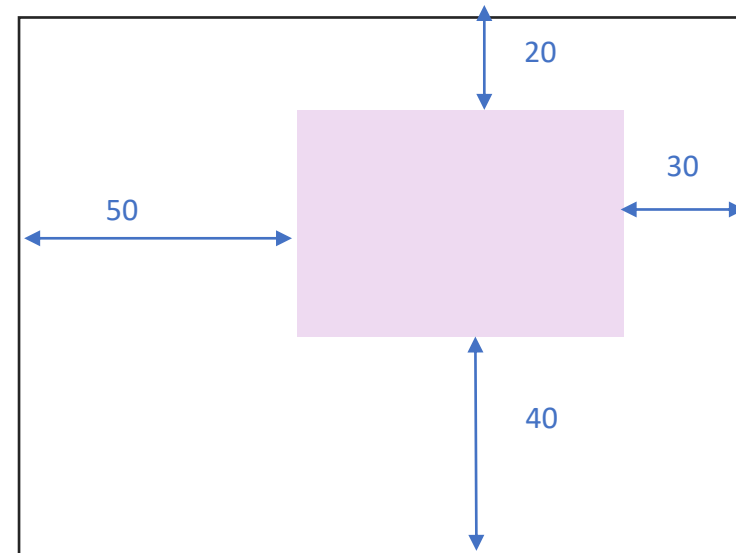
The **box model** attributes



```
p {  
  margin: 20px 30px 40px 50px ;  
}
```

Diagram illustrating the margin values for a box model:

- TOP: 20px
- RIGHT: 30px
- BOTTOM: 40px
- LEFT: 50px





10 MIN

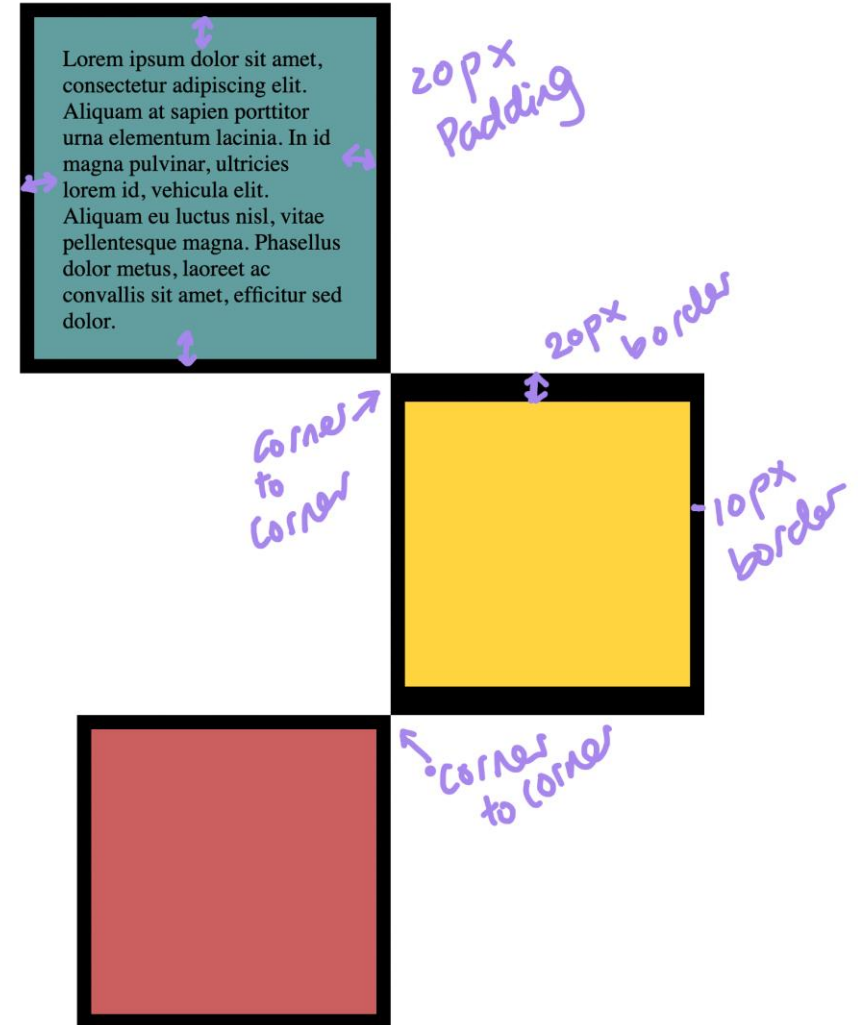


Activity 2

Create 3 Boxes following given constraints:

CONSTRAINTS

- Divs width and height : 200px
- Divs Background color : follow mockup
- **Divs shall be displayed corner to corner**
- Div 1 : padding 20px, border 10px
- Div 2 : top and bottom border 20px, left and right border 10px
- Div 3 : border of 10px





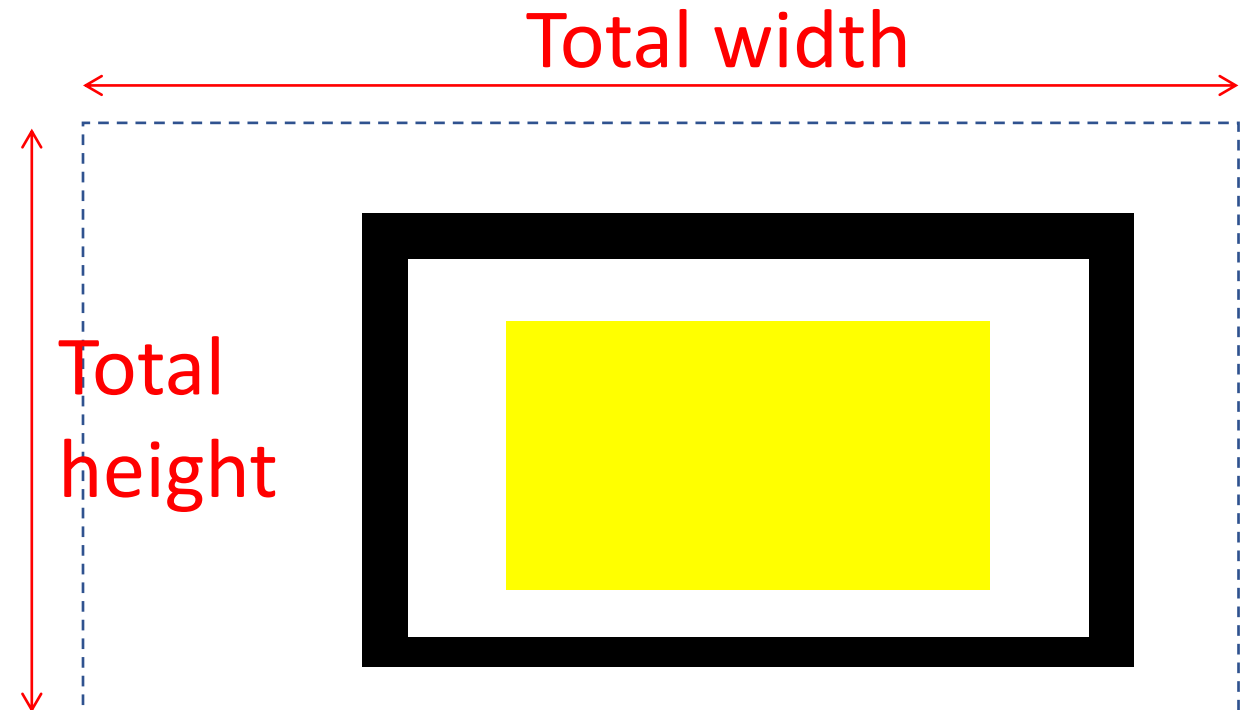
10 MIN

Calculate an element total size

To calculate the full size of an element, you must take into account

- height width
- paddings
- margins
- borders

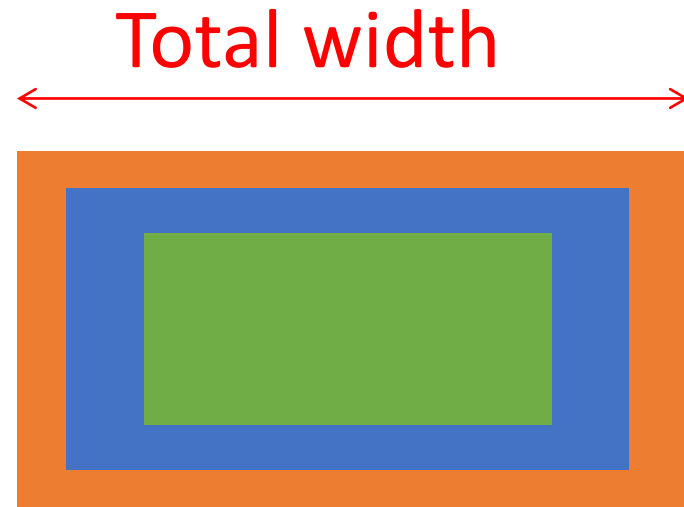
```
div {  
  margin : 20px 20px 10px 50px;  
  border : 10px solid black;  
  padding : 20px;  
  height : 100px;  
  width : 400px  
}
```



Compute the total width and height, given the above code

What will the **total width** of this element ?

```
.my-box {  
  width: 200px;  
  border: 10px solid;  
  padding: 20px;  
}
```



A

200px

B

220px

C

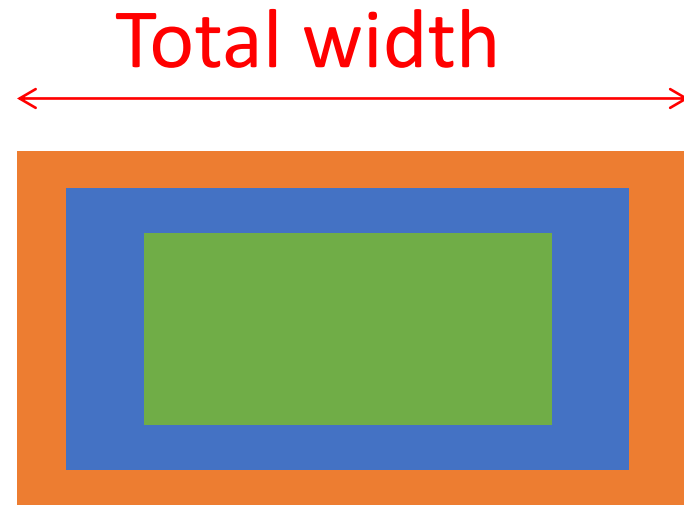
230px

D

260px

What will the **total width** of this element ?

```
.my-box {  
  width: 200px;  
  border: 10px solid;  
  padding: 20px;  
}
```



A

200px

B

220px

C

230px

D

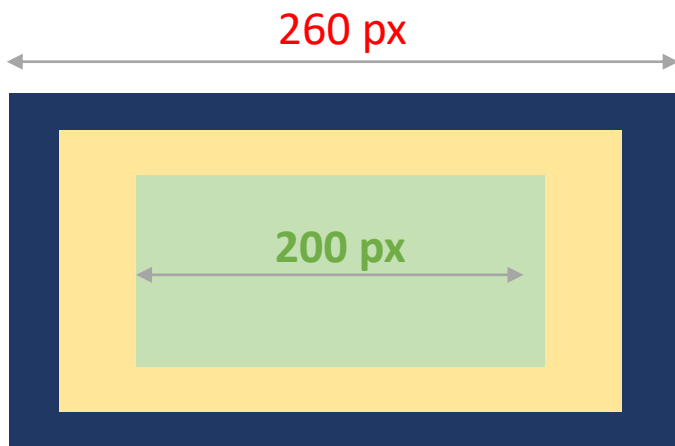
260px

$200 + 2 * 10 + 2 * 20$

What is Box Sizing ?

Box sizing defines from where the **browser** is going to **measure** an objects size

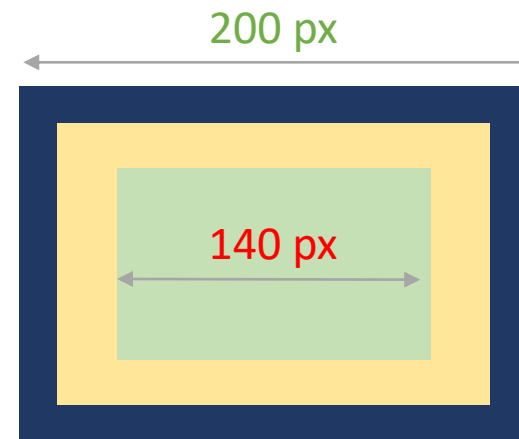
content-box (default value)



width: 200px;
padding: 20px;
border: 10px;
box-sizing: **content-box**;

- Total width = **260px** ($200\text{px} + 20\text{px} + 20\text{px} + 10\text{px} + 10\text{px}$)
- Content width = **200px**

border-box



width: 200px;
padding: 20px;
border: 10px;
box-sizing: **border-box**;

It's sometime easier to include padding and border when specifying the element dimension!

- Total width = **200px**
- Content width = **140px** ($200\text{px} - 20\text{px} - 20\text{px} - 10\text{px} - 10\text{px}$)



CSS Length Units

EM vs. PX vs. PT vs. CM vs. IN vs. REM



CSS Units

Percentage based

%

Relative to the value of parent element. 100% is the width of the parent element

Font based

em

Relative to the font-size of the parent element.

Viewport based

vh

equal to 1% of the height of the browser window size.

px

Pretty self explanatory .Absolute length in pixel

rem

Relative to font-size of the root element.

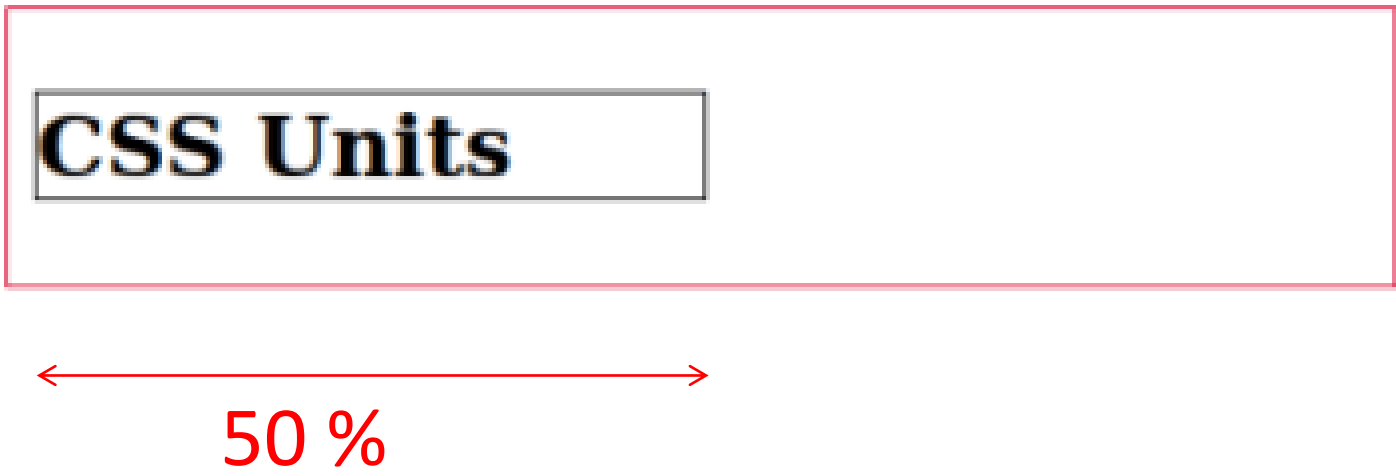
vw

equal to 1% of the width of the browser window size.

Percentage based unit

The length is **relative** to its immediate parent element size.

```
.box{  
  width: 500px;  
  border: 1px solid crimson;  
  padding: 10px;  
}  
h1{  
  width: 50%;  
  border: 1px solid;  
}
```



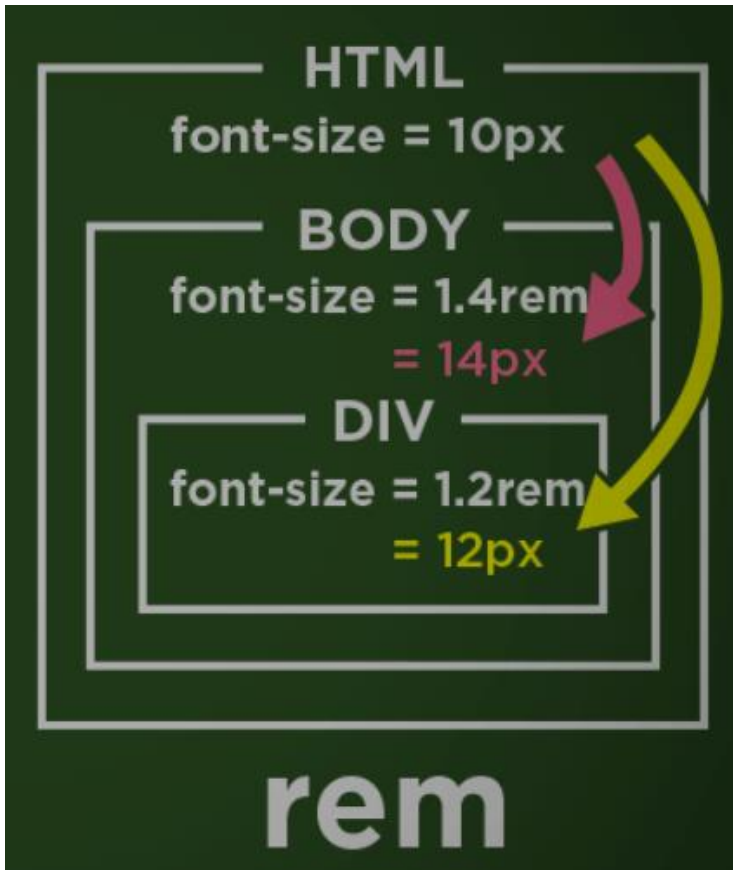
CSS Units

50 %

Font based units

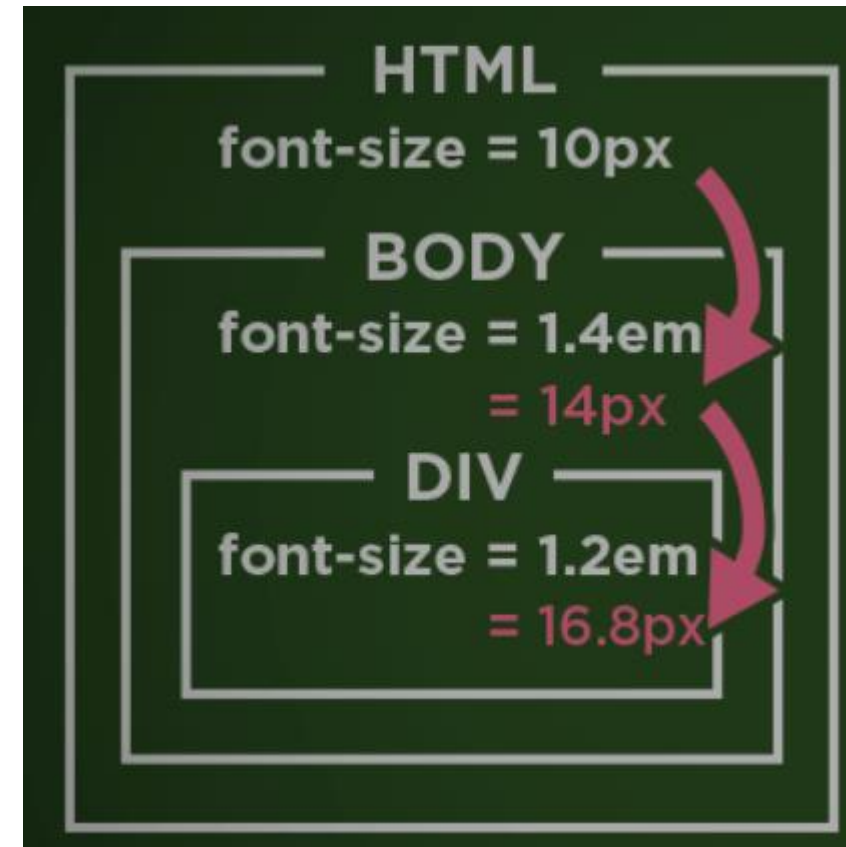
rem unit

The length is **relative** to the **root** element **font size**



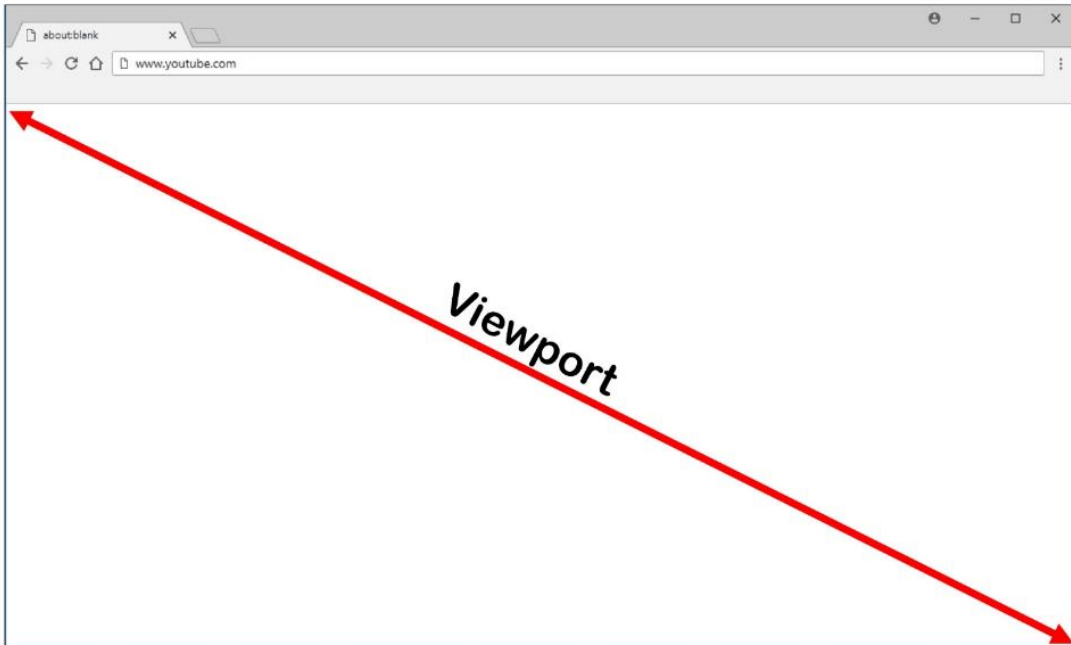
em unit

The length is **relative** to the **parent** element **font size**



Viewport based units (vh, vw)

The viewport is the **user's visible area** of a web page



```
div {  
  height: 100vh  
}
```

100vh is equal to 100% of the height of the viewport

```
div {  
  width: 50vw  
}
```

50 vw is equal to 50% of the width of the viewport



Research Assessment

Before session **C3-S4**, make researches regarding the bellow topics:

TOPIC 1 - when should we use **absolute units** ?

TOPIC 2 - when should we use **font based units** ?

TOPIC 3 - when should we use **viewport units** ?

Kittens heading px

Cat ipsum, good morning sunshine.
Wake up human for food at 4am
intently stare, this human feeds me.

These paragraphs are using **font-size: 16px, line-height: 24px**

Kittens heading rem

Cat ipsum, good morning sunshine. Wake up human
for food at 4am intently
stare, this human feeds
me.

These paragraphs are
using **font-size: 1rem, line-height: 24px**

Kittens heading rem

Cat ipsum, good morning
sunshine. Wake up human
for food at 4am intently
stare, this human feeds
me.

These paragraphs are
using **font-size: 1rem, line-height: 1.5**

Behavior of units when zooming

Expected outcome

- Different uses cases where each unit is more appropriate
- Illustration of each case with some code

Tips

- Consider accessibility, multiple device, responsivity etc....



Congratulations !



You should now master those concepts

BOX MODEL

- ✓ Make the difference between **block** and **inline** elements
- ✓ Understand and manipulate the **Box Model** (Margin, Padding Border)
- ✓ Understand the **Box Sizing** modes to compute object dimensions

UNITS

- ✓ What are the **main types of units** in web design

3-2-1 Challenge

- ✓ List three things you **learned** today.
- ✓ List two **questions** you still have.
- ✓ List one aspect of the lesson or topic you **enjoyed**.



After this session

1 - Review the theory

- ✓ Review the [Box Model](#)
- ✓ Review [CSS units](#)

2 - Work on your Research assessment

- ✓ Presentations on C3-S4 !

